

The Cosmological Constant as the Count-Conjugate of a Finite Null-Edge Spacetime: a kernel-checked everpresent- Λ core and its pre-registered dichotomy

[named authors — release gate]

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Abstract

We report a Lean 4 formalization, over Mathlib, of the arithmetic core of a finite “null-edge” account of the cosmological constant Λ . The value proposition is exactness and machine-verification of the *structural* claims, not a first-principles derivation of the observed value. Three independently landed, kernel-checked (**M**) results organize the account. (1) *Magnitude dissolution, structurally*: in the finite spectral-action grading the coefficient of $\text{tr}(\mathbf{1})$ is invariant under *every* deformation of the Dirac operator, while the order-2 and order-4 traces provably do change; only count statistics can touch the order-0 coefficient. (This grading invariance is the finite avatar of a classical heat-kernel fact; the physical statement that no matter or gauge channel renormalizes Λ rests on the finite unimodular trade of Section 4, not on the grading alone — see the Section 4 remark.) (2) *The everpresent scaling law*: with the discreteness (Poisson) input $\langle \delta N^2 \rangle = V$ the normalized $\Lambda = \delta N/V$ has RMS $1/\sqrt{V}$, routed through the framework’s own extensive edge count rather than an abstract volume. (3) *A pre-registered dichotomy with a sharp exponent*: the count variance $\text{Var}(N)$ is a finite susceptibility whose scaling exponent α decides everpresent survival — the Poisson branch ($\alpha = 1$) gives the everpresent exponent $-\frac{1}{2}$ and predicts fluctuating dark energy at $1/\sqrt{V}$, while any hyperuniform branch ($\alpha < 1$, the constraint-correlated case realized by Coulomb systems) suppresses Λ below $1/\sqrt{V}$ and kills the everpresent identification. This dichotomy is now a **M** theorem on the framework’s own fermionic states: via Wick’s number-variance identity, an explicit projection (Fermi-sea) kernel realizes a genuinely sub-extensive, *unbounded* variance ($\alpha = \frac{1}{2}$), so the fork is proved and only the physical count-identification remains open. We additionally give an *illustrative* finite Fourier/uncertainty picture of the Λ -volume conjugacy on a small register, and pre-register the canonical finite Henneaux–Teitelboim statement ($\{\Lambda, V\} = 1$, $\dot{\Lambda} = 0$ on shell) as its proper successor. Every trust boundary is stated at its point of use. What is *not* claimed: a derivation of the value 10^{-122} (the 4-volume V is an input), of the sign, or of the stochastic dynamics. As of this draft the capstones are kernel-clean *and* build-guarded: a dedicated axiom guard (**LambdaCosmologyAxiomGuard**) pins the footprint of every cited theorem on every build (Section 7).

Claim calculus. **M** = a theorem kernel-checked in this repository (axiom footprint `{propext, Classical.choice, ...}`, every module cited below is **native_decide**-free). **[IMPORT]** = an external result relied on but not proved here. **[C]** = a pre-registered conjecture with a stated kill condition. **[SPEC]** = honest speculation. Unmarked prose is interpretation.

1 The problem, in two halves

Magnitude. Continuum QFT’s naive vacuum zero-point density is $\sim M_{\text{Pl}}^4$, about 120 orders above the observed $\rho_\Lambda \sim (10^{-3} \text{ eV})^4 \sim 10^{-122}$ in Planck units. *Coincidence.* Why is ρ_Λ comparable to

the matter density *now*, when the two scale differently with expansion? A finite theory has no divergent zero-point sum, so the magnitude problem cannot appear in its continuum form; but finiteness alone does not *predict* smallness, and matter loops still renormalize the vacuum energy (Weinberg 1989 [IMPORT]). The framework’s genuine leverage is (i) a *structural* reason no channel can feed the order-0 coefficient (Section 4), and (ii) a specific, falsifiable handle on the coincidence half (Sections 3,5).

2 The finite spectral grading and the order-0 coefficient

In the spectral-action route (Chamseddine–Connes–Marcolli [IMPORT]) the action is one functional whose expansion gives a cosmological/volume term (order 0), a gravity term (order 2), and matter terms (order 4). Thus Λ is the order-0 coefficient $a_0 \text{tr}(\mathbf{1})$ — a finite, computable number, and a graded piece of the same functional yielding gravity and matter.

The finite avatar makes this exact. In `LambdaMomentHierarchy` the graded pieces are explicit and the following hold `M`:

- `moment_hierarchy`: the order-0/2/4 decomposition of the finite functional into $(a_0 \text{tr} \mathbf{1}, \text{gravity}, \text{matter})$;
- `order0_deformation_invariant`: $\text{order0}(a_0, D + P) = \text{order0}(a_0, D)$ for *any* perturbation P ;
- `only_count_touches_lambda` and `hierarchy_verdict`: the order-0 coefficient is blind to every channel deformation, the higher orders are not.

3 The main handle: everpresent Λ

Mechanism (Sorkin; Ahmed–Dodelson–Greene–Sorkin [import]). In unimodular gravity Λ is conjugate to the spacetime 4-volume V ; in a discrete spacetime the element count $N \sim V$, a Poisson fluctuation gives $\delta N \sim \sqrt{V}$, and conjugacy makes Λ fluctuate about zero with $|\Lambda| \sim \delta N/V \sim 1/\sqrt{V}$. For the present horizon $V \sim 10^{244}$ Planck units, so $1/\sqrt{V} \sim 10^{-122}$ — the observed magnitude, untuned [IMPORT] — and since $\Lambda \sim 1/\sqrt{V}$ tracks the ambient density at every epoch, the coincidence is dissolved.

Kernel core. The scaling arithmetic is machine-checked, first abstractly and then through the framework’s own primitive:

- `everpresentLambda_secondMoment_eq_inv_volume`, `everpresentLambda_rms_eq_inv_sqrt_volume` `M`: given $\langle \delta N^2 \rangle = V$, $\Lambda = \delta N/V$ has second moment $1/V$ and RMS $1/\sqrt{V}$ (`NullEdgeP9EverpresentLambdaScale`);
- `edgecount_extensive`, `edgecount_mono`, `lambda_rms_eq_inv_sqrt_count`, `everpresent_verdict` `M`: the pierced-null-edge count N is additive over disjoint regions and monotone, and the everpresent RMS holds with N *the edge count* (`LambdaEdgeCount`) — the handle is native, not borrowed via an abstract V .
- `uniformSecondMoment_antitone`, `uniformSuppression_below_everpresent` `M`: a uniform suppression with residual A^2/N beats the $\sqrt{N}$ fluctuation iff the scale is sub-extensive (`NullEdgeP9EverpresentLambdaScale`) — the comparison machinery for Section 5.

Honest scope. `M` is the *arithmetic of the scaling*. The inputs remain [IMPORT]/[C]: the Λ – V conjugacy (unimodular gravity), the Poisson second moment $\langle \delta N^2 \rangle = V$ (Section 5 situates this), and the identification of the edge count with the 4-volume.

4 Structural magnitude dissolution (the headline theorem)

The order-0 blindness of Section 2 upgrades the “finiteness removes the divergence” story to a structural statement: matter physics *cannot* renormalize Λ . In `LambdaUnimodular` and the capstones:

- `trace_one_eq_dim` ($\text{tr } \mathbf{1} = n$), `trace_channel_blind`, `order0_operator_blind` **M**: the order-0 coefficient is invariant under channel and operator deformations.
- `multiplier_field_equation`, `vacuum_shift_is_gauge`, `gauge_solution_map` **M**: the finite unimodular trade — under a count constraint a shift of the order-0 coefficient is a gauge move on the constraint surface (unobservable), the multiplier enters the field equation as $+\Lambda \mathbf{1}$ (Jacobson’s integration constant, finitely), and the observable residue is the count’s deviation from its constrained mean.
- `lambda_only_count_can_move_order0` **M** (`LambdaMagnitudeCapstone`): bundles it — $\text{order0}(D+P) = \text{order0}(D)$ and Λ invariant under vacuum shifts, *while* $\text{tr}((D+P)^2) \neq \text{tr}(D^2)$ and the order-4 trace likewise moves. Matter physics moves the gravity and matter sectors and leaves Λ untouched; only count statistics can move it.
- `lambda_magnitude_capstone`, `lambda_nonvacuity_witnesses` **M**: the magnitude verdict with explicit non-vacuity witnesses.

Remark 1 (what this is and is not, and where the load-bearing wall really is). *The order-0 invariance is the finite avatar of a classical heat-kernel fact — the leading Seeley–DeWitt coefficient a_0 depends only on rank $\times$ volume, not on the operator [IMPORT] — so order0 not mentioning D makes `order0_deformation_invariant` close to definitional (it is the structural encoding of the grading, best read as the Classical-analogue it is). Consequently the grading theorem alone does not carry the physical claim “matter cannot renormalize Λ ”: as Section 1’s Weinberg concession already admits, a Higgs-sector vacuum shift sits in the order-4 trace and still gravitates as an effective Λ . What neutralizes it is the finite unimodular trade — `vacuum_shift_is_gauge` together with the imported Λ – V conjugacy (Section 3): under a fixed-count constraint a uniform $c\mathbf{1}$ shift is a gauge move on the constraint surface, hence unobservable. That trade, not the grading, is the load-bearing wall, and the paper should be read that way. This dissolves the magnitude problem structurally for the finite functional but does not derive the value ($\text{tr } \mathbf{1} = (\text{edge count}) \times (\text{fibre dimension})$, V an input). The honest finite sequestering core still owes two keystones, both landable from `LambdaUnimodular` (pre-registered, not yet theorems): (i) an ensemble shift-invariance — at fixed count the partition function and every observable are invariant under a uniform $c\mathbf{1}$ shift, upgrading the trade from classical field equation to statistical mechanics; and (ii) a local-shift decomposition — an inhomogeneous $c(x)\mathbf{1}$ splits into a gauged uniform mode and a residue provably routed into the order-2/4 observables.*

5 The pre-registered dichotomy and the exponent prediction

The Poisson input need not be assumed: it is a statement about the count variance, which the finite ensemble computes.

Λ as a susceptibility. `LambdaSusceptibility` **M**: `expect_Ncount`, `var_count` give $\text{Var}(N)$ for independent occupancies, and `bernoulli_bound` proves $\text{Var}(N) = \sum_i p_i(1 - p_i) \leq \langle N \rangle$ (equality in the sparse/Poisson limit). Everpresent becomes an *upper-bound theorem for an ideal edge gas*: $\Lambda_{\text{rms}} \leq 1/\sqrt{\langle N \rangle}$, with $\Lambda_{\text{rms}} = \sqrt{\chi_N}/\langle N \rangle$ the count susceptibility.

The fork [C](the distinctive question). Do the framework’s own constraints correlate the count? Constraint-induced correlations can make number fluctuations sub-extensive — the physical precedent is that Coulomb systems are hyperuniform (charge fluctuations scale with surface, not volume) [IMPORT]. `LambdaCountDichotomy M` formalizes both branches on finite witnesses: `free_variance_extensive/free_extensive` (independent edges, $\text{Var} \sim N$, Poisson) versus `constrained_variance_hard/hard_subextensive` and `soft_variance/soft_subextensive` (a constrained witness with strictly sub-extensive variance). What the kernel currently proves is that both branches are *possible* on witnesses — which nobody doubted; the answer, not the possibility, is the paper, and it is computable because the counted objects are fermionic.

Resolving the fork on the framework’s own fermionic states [C] (the decisive upgrade).

For any free-fermion/Gaussian (CAR) state with correlation matrix K ($0 \leq K \leq 1$), Wick’s theorem gives the regional number variance in one line, $\text{Var}(N_A) = \text{tr } K_A - \text{tr } K_A^2$; the diagonal case $K = \text{diag } p$ recovers `var_count` above (the extensive branch). A *projection* kernel $K^2 = K$ — a Fermi-sea, i.e. *kinematic*, count — instead makes $\text{Var}(N_A)$ count only boundary-crossing modes. An explicit finite family (b fully-occupied bulk sites plus k boundary bonds) has $\text{Var} = k/4$ with region size tunable to k^2 : a genuinely *sub-extensive and unbounded* variance, $\alpha = \frac{1}{2}$ — the non-degenerate hyperuniform witness the kill branch needs, on a bona fide kernel rather than a degenerate toy. The fork then sharpens into a *dichotomy*: *everpresent Λ survives iff the Λ -conjugate count is the thermal/diagonal count rather than the kinematic/projection (vacuum-weighted) count*. Two identifiable results tension it from both sides: the framework’s own vacuum (finite Sorkin–Johnston / discrete free-fermion) reading is generically sub-extensive (Giovè–Klich regime) and would fire the kill, while a Lorentz-invariance leg (Bombelli–Henson–Sorkin: Poisson is the essentially unique Lorentz-invariant discretization [IMPORT]) pins the *kinematic sprinkling* count at $\alpha = 1$. The *mathematical* dichotomy is now `M` (`LambdaFermionicFork`, guard-pinned): the fermionic number-variance identity, an explicit boundary-bond *projection* kernel ($K^2 = K$, kernel-checked — a genuine Fermi-sea correlation matrix, `bondProj_isProjection`) with $\text{Var} = k/4$ exactly (`bondProj_numberVariance`) and region size tunable to k^2 (`fork_subextensive`: $\alpha = \frac{1}{2}$, sub-extensive *and* unbounded), and the two-branch verdict (`fermionic_fork_verdict`). So the fork is a theorem, not merely a possibility: a diagonal/thermal count is extensive and everpresent survives, while a projection/kinematic count can be genuinely sub-extensive and everpresent is killed — on a bona fide kernel, not a degenerate toy. What remains [C] is only the *physical* count-identification: which count the framework’s own dynamics makes Λ -conjugate (bare vs vacuum-weighted). Either resolution publishes; a hyperuniform no-go for everpresent Λ in constrained fermionic discreta would be the higher-impact outcome.

The exponent, exactly. `LambdaExponentFork M` turns the fork into a single exponent. With $\text{Var}(N) \sim N^\alpha$ the Λ exponent is

$$\text{lamExp}(\alpha) = \frac{\alpha}{2} - 1 \quad (\text{lamExp_closed}),$$

so `everpresent_value`: $\text{lamExp}(1) = -\frac{1}{2}$ (Poisson); `hyperuniform_faster`: $\alpha < 1 \Rightarrow \text{exponent} < -\frac{1}{2}$ (suppressed below $1/\sqrt{V}$; e.g. area scaling $\alpha = \frac{3}{4}$ gives $\sim 10^{-152}$); `superextensive_slower`; and `fork_iff`: $\text{lamExp}(\alpha) = -\frac{1}{2} \iff \alpha = 1$. In program discipline the prediction is an *exponent*, not a scale: a measured deviation of the Λ -variance exponent from $-\frac{1}{2}$ measures hyperuniformity.

Kill condition [C]. If the framework’s Gauss/gauge constraints force $\alpha < 1$ on physical witnesses, the everpresent identification of observed dark energy dies in this framework and Λ demotes

to an unexplained integration constant. Which count is conjugate to Λ (bare edge count vs decorated/soldered count) is decidable on finite witnesses; either answer is publishable.

6 A finite Fourier/uncertainty picture of the conjugacy (illustrative)

`LambdaConjugacy M` builds a volume register and a Λ partner by a finite DFT over $\mathbb{Z}/4$ and proves a support-uncertainty relation: `delta_maps_to_uniform` and `uniform_maps_to_delta` (a sharp count is a flat Λ and conversely), `support_uncertainty` (a Donoho–Stark support bound), `delta_saturates` (the extremal register saturates it). We are careful not to oversell this: the $\mathbb{Z}/4$ support-uncertainty relation is an *illustrative* finite avatar of “ Λ conjugate to volume”, a metaphor-made-exact on one small register, *not* the canonical conjugacy statement. The dictionary it suggests — mass = residual mixedness of hidden null *directions*, Λ = residual phase noise of hidden null *volumes* (the same trace-out-the-hidden-register operation on two registers) — is interpretation, not theorem. Two successors make “native” literal: (i) the general- N Donoho–Stark support bound $N \leq |\text{supp } f| \cdot |\text{supp } \hat{f}|$ over $\mathbb{Z}/N$, now `M` for *all* N (`LambdaUncertaintyGeneralN.support_uncertainty`, via a machine-checked Plancherel and Cauchy–Schwarz on Mathlib’s `ZMod.dft`; guard-pinned), which retires the “ $\mathbb{Z}/4$ witness only” scope of this section — its extremizer classification is the remaining piece; and (ii), the real prize, a finite Henneaux–Teitelboim statement — a constrained Hamiltonian minisuperspace in which Λ is the momentum conjugate to total 4-volume, $\{\Lambda, V\} = 1$, $\dot{\Lambda} = 0$ on shell, with `vacuum_shift_is_gauge` as a corollary. To our knowledge unimodular gravity’s Λ -as-integration-constant has not been machine-checked even in a minisuperspace avatar; that theorem, not the DFT witness, is what would make the conjugacy native. (Both are [\[C\]](#)/pre-registered.)

7 Verification status, and the one honest gap that remains

Every module cited is kernel-clean (`native_decide-free`; footprint `{propext, Classical.choice, Quot.sound}`) and imported in the draft build root.

1. **Build-enforced axiom guard: in place.** Every headline theorem cited above is now pinned by a `#guard_msgs/#print axioms` block in `PhysicsSM/Draft/NullEdge/LambdaCosmologyAxiomGuard.lean` (about forty pins across the eleven Λ modules; build green). The kernel-clean footprint is therefore re-checked on *every* build: should any dependency ever regress — a `sorry`, an `axiom`, or a `native_decide` entering a transitive proof — the recorded axiom string stops matching and the guard module fails to build. A single pin is instructive: `edgecount_mono` is guarded at the sharper footprint `{propext, Quot.sound}` (it needs no choice), so even a spurious `Classical.choice` would trip the guard. (Pattern: the sibling `OvernightTheoryAxiomGuard`.)
2. **This document is the surfacing (the remaining gate).** Until now the results lived only in draft modules; this manuscript is rung L6. It requires named authors and a primary-source pass on every `[IMPORT]` before leaving draft.

8 The event horizon (what is *not* claimed)

No derivation of V , hence not of the value 10^{-122} : V (the age/size of the universe) is an input. No derivation of the *sign* (the positive-sector/Krein bias is a one-line [\[SPEC\]](#) question at most). No claim on the stochastic *dynamics* (sequential growth is causal-set theory’s, not ours). The claim

to defend is exactly: the framework *dissolves the magnitude divergence structurally* (Sections 2,4), *explains the coincidence conditionally* on the Poisson branch (Section 5), and derives neither the value nor the sign.

9 Predictions (conditional on the Poisson branch) [C]

1. $\langle \Lambda \rangle \approx 0$, $\Lambda_{\text{rms}} \sim V^{-1/2}$; the observed positive value is a realization, the sign undetermined.
2. Coincidence is *permanent*: $|\rho_\Lambda| \sim \rho_{\text{crit}}$ at every epoch, tiny because the universe is old and large.
3. Deviations are *horizon-scale* (low-frequency drift), not a clustering local fluid.
4. CPL (w_0, w_a) fits may *mislead*: stochastic $\Lambda(t)$ can fake smooth evolution; the object is volume noise, not a rolling scalar.
5. No exact future de Sitter equilibrium: $\Lambda_{\text{rms}} \sim t^{-2}$ keeps tracking; long-term acceleration may be intermittent.

Inherited caveat (out of claimed scope). Predictions 1 and 5 lean on temporal structure this paper does not supply: a zero-mean Λ that is observed *persistently* positive over $\sim$ Hubble times requires long correlation times in the stochastic history, which is Zwane–Afshordi–Sorkin territory (the dynamics of everpresent Λ), not the finite kinematic core formalized here. We flag rather than assume it; a genuine prediction of the sign’s persistence would need that dynamical input.

Observational posture [import] (verify before citing). Planck 2018 sets rigid Λ CDM as baseline; DESI DR2 BAO+CMB report a time-evolving equation of state at $\sim 3.1\sigma$ (2.8–4.2 σ with supernovae) — *hints*, not a replacement. The framework inherits a falsifiable phenomenology (fluctuating, sign-changing, $1/\sqrt{V}$ -tracking dark energy) *conditional on the Poisson branch*. Live tension: Zwane–Afshordi–Sorkin (everpresent models can fit as well as Λ CDM and relieve low- z tensions) vs the Aspects-II finding (CMB-favored fluctuation magnitudes smaller than typical everpresent models). A confirmed rigid $w = -1$ with tight fluctuation bounds is the kill; a confirmed deviation is friendly but not a confirmation.

Landed-theorem table

Statement	Lean declaration	Grade
order-0 coefficient invariant under any deformation; higher orders not	<code>LambdaMomentHierarchy.order0.deformation_invariant,</code> <code>only_count_touces_lambda</code>	M
magnitude verdict: only count statistics move Λ	<code>LambdaMagnitudeCapstone.lambda_only_count_can_move_order0</code>	M
finite unimodular trade / multiplier / gauge shift	<code>LambdaUnimodular.multiplier_field_equation,</code> <code>vacuum_shift_is_gauge</code>	M
everpresent scaling $\Lambda_{\text{rms}} = 1/\sqrt{V}$	<code>...EverpresentLambdaScaling.everpresentLambda_rms.eq_inv_sqrtV</code>	M
routed through the native edge count	<code>LambdaEdgeCount.lambda_rms.eq_inv_sqrtVcount,</code> <code>edgecount_extensive</code>	M
Λ as susceptibility; Bernoulli bound $\text{Var}(N) \leq \langle N \rangle$	<code>LambdaSusceptibility.var_count,</code> <code>bernoulli_bound</code>	M
Poisson vs hyperuniform dichotomy on finite witnesses	<code>LambdaCountDichotomy.free_extensive,</code> <code>hard_subextensive</code>	M
exponent law $\text{lamExp}(\alpha) = \alpha/2 - 1$; $\text{lamExp}(1) = -1/2$; fork iff	<code>LambdaExponentFork.lamExp_closed,</code> <code>everpresent_value, fork_iff</code>	M
finite Λ -volume conjugacy & support uncertainty	<code>LambdaConjugacy.support_uncertainty,</code> <code>delta_saturates</code>	M
$1/\sqrt{V} \sim 10^{-122}$ for the present horizon	Ahmed–Dodelson–Greene–Sorkin	[IMPORT]
Λ - V conjugacy (unimodular gravity)	Henneaux–Teitelboim; Unruh	[IMPORT]
Λ = order-0 spectral-action term	Chamseddine–Connes–Marcolli	[IMPORT]
which count is conjugate; kill on $\alpha < 1$	—	[C]
value 10^{-122} , sign, stochastic dynamics	—	outside

References (verify before any manuscript citation) [\[import\]](#)

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